

Statistics reveal great improvements in domestic robots

The worldwide number of domestic robots will rise to 31 million by the end of 2019. Sales volumes of robots that clean floors, mow lawns and clean swimming pools will grow to about thirteen billion dollars.

In the coming years, vacuum and floor-cleaning robots will continue to make up the lion's share of units at work in households. Sales will rise to 3.6 million units within the 2016-2019 forecast period. Vacuum and floor-cleaning robots account for 96 per cent of total domestic robot sales.

Robotic grass cutters and pool-cleaning robots rank second and third, respectively. Sales figure of assistive robots for handicapped and elderly people is expected to rise to 37,500 units from 2016 to 2019.

Robots have become advanced enough to penetrate homes. The market for consumer robots has grown at a CAGR of seventeen per cent between 2014 and 2019.

Robots are slowly entering Indian homes, too. For example, Silverlit robot, priced at ₹ 1875, can go after a ball and kick it, walk around the room without bumping into obstacles—it can even hold objects. Euroclean and RoboCleanz from Eureka Forbes, priced at ₹ 6690, can clean floors, safely wandering around the house.

By 2019, 42 million service robots will operate for domestic use. Japan is expected to have more social robots due to its high aging population. Personal robots are set to be an 18.85-billion-dollar opportunity by 2020. The Asian market for domestic robots will explode with China emerging as one of the major countries expected to provide innovative, low-cost, autonomous domestic robots in the next few years.

Robots in space

Space astronauts frequently face several issues, during their long-term stays in International Space Station. "For long-term space missions where communication with friends and families becomes difficult, PARO can prove to be a great companion," opines Japanese industrial automation company AIST.

Japan has sent the world's first talking social robot astronaut, Kirobo, to

space. "Russia was the first to reach the moon, but Japan is the first to send a robot astronaut with the ability to communicate with humans in space," says Yorichika Nishijima, project manager for Kirobo.

Kirobo is about 34cm tall and weighs about one kilogram. It was sent from Tanegashima Space Center on August 2013. Kirobo has broken two Guinness World Records after it returned to Japan, following its eighteen-month stay on International Space Station—first companion robot in space and highest altitude for a robot to have a conversation.

More opportunities in the future

There are many other trends catching up in the home robotics sector. For example, open source software and hardware are greatly helping those enthusiastic about home robotics, including hobbyists. With the help of these tools, they can assemble home robots at a low cost and program them according to their requirements.

Even commercial robotic companies through their R&D department are trying to develop smart robots to make domestic robots affordable. Soon, domestic robots could be available in the market at the cost of a cellphone.

Cloud-connected home robots are already becoming part of our lives. Multi-functional robotic chefs can fry, steam, bake and perform other actions without human intervention. These cloud-connected robots are likely to evolve into more advanced versions in the near future.

Owing to such technologies as AI, the Internet of Things (IoT), Big Data and others, home robots can carry out a variety of human tasks with high precision. The rise of AI technologies such as machine learning, computer vision, natural language processing and gesture controls is changing the landscape of domestic robots.

Cloud robots, 3D vision, virtual foresight and smart materials are other technologies that can help further expand the home robotics market. It is expected that embedded devices too will make these smarter. And, since technology is the main differentiator, home automation is mainly achieved using home robots. **EFY**

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